BENCHMARK REPORT PubTabNet Dataset Evaluation

# Table Extraction Benchmark & Performance Analysis

Evaluating three extraction pipelines across structural accuracy, character recognition, and runtime speed on 50 PubTabNet samples.

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PRIMARY GOAL
**Evaluate extraction quality & structure**

CORE METRICS
**TEDS, TEDS-Struct, CER, Runtime**

DECISION TARGET
**Accuracy vs Speed trade-offs**

PIPELINE ARCHITECTURE

# Three Evaluated Extraction Approaches

METHOD 1

## PyMuPDF4LLM forced OCR

Applies OCR directly via PyMuPDF4LLM engine.

Converts extracted content directly into target table structure.

METHOD 2

## Tesseract OCR + PyMuPDF4LLM

Generates a searchable OCR PDF using Tesseract.

Processes resulting searchable layer via PyMuPDF4LLM parser.

METHOD 3

## MinerU standard

Runs standard end-to-end document parsing pipeline.

Native layout model output evaluated against references.

ENVIRONMENT

# Dataset Configuration & Experimental Setup

DATASET SOURCE

**nhhsag12/pubtabnet-with-html**

IMAGE PREPROCESSING

**2× Upscaling Applied**

LATENCY TRACKING

**Seconds per sample (s/sample)**

**50**

**Evaluated Samples**

0 skipped • 100% evaluated against ground truth annotations

EVALUATION METRICS

# Quantitative Benchmark Metrics

## TEDS (Full)

Range: 0–1 ↑

Tree-Edit-Distance-based Similarity measuring structural alignment and cell text precision together.

## TEDS-Struct

Range: 0–1 ↑

Isolated table structure similarity ignoring cell text contents entirely.

## CER (Character Error Rate)

Lower is Better ↓

Character recognition error rate across cells.

$$\mathrm{CER} = \frac{S + D + I}{N}$$

## Runtime Latency

Lower is Better ↓

Average end-to-end processing duration measured in seconds per sample.

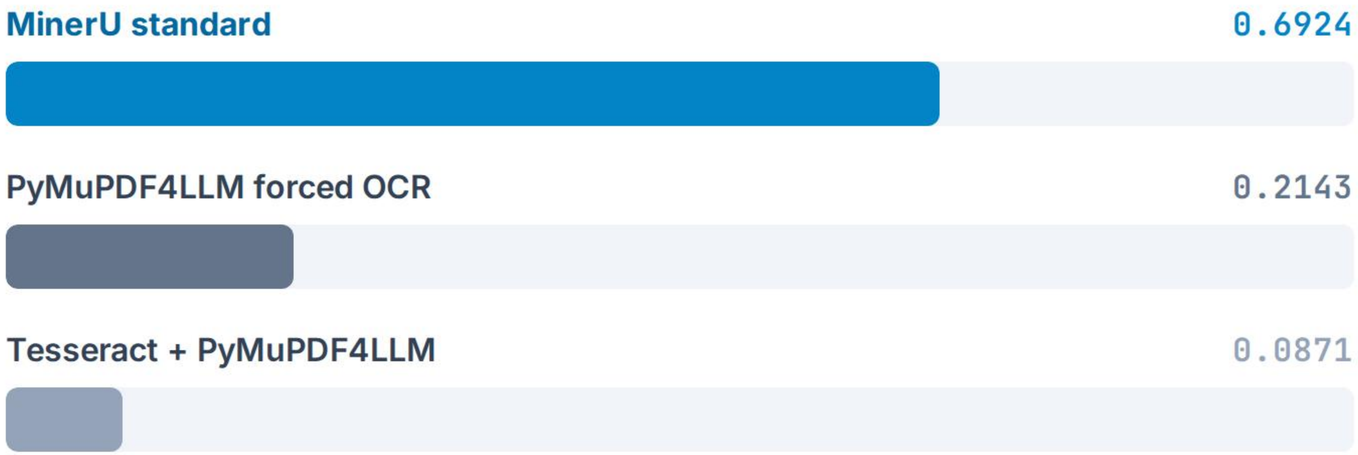
SUMMARY RESULTS

# Comprehensive Benchmark Results

| Method | TEDS ↑ | TEDS-Struct ↑ | CER ↓ | Seconds/sample ↓ |
|---|---|---|---|---|
| PyMuPDF4LLM forced OCR | 0.2143 | 0.3001 | 0.6668 | 0.4737 |
| Tesseract + PyMuPDF4LLM | 0.0871 | 0.1491 | 0.7524 | **0.3731** |
| **MinerU standard** | **0.6924** | **0.7389** | **0.1644** | 5.7057 |

ACCURACY ANALYSIS

# TEDS Overall Extraction Quality




## 3.2× Higher

MinerU TEDS score vs direct PyMuPDF4LLM forced OCR.

## 8.0× Higher

MinerU TEDS score vs Tesseract OCR-PDF pipeline.

STRUCTURAL ANALYSIS

# Table Structure Alignment (TEDS-Struct)

MINERU STANDARD

**0.7389**

High Structural Integrity

FORCED OCR

**0.3001**

Frequent Row/Col Shift

TESSERACT + PYMUPDF

**0.1491**

Severe Grid Breakdown

⚠ Structural reconstruction failure is the primary bottleneck for standard OCR text-layer parsers.

RECOGNITION ACCURACY

# Character Error Rate (CER) Comparison

LOWEST ERRORS

## MinerU Standard

**0.1644**

Robust text extraction with minimal deletion or substitution noise.

MODERATE ERRORS

## PyMuPDF4LLM forced OCR

**0.6668**

High character-level distortion across dense cell content.

HIGHEST ERRORS

## Tesseract + PyMuPDF4LLM

**0.7524**

OCR PDF conversion layer creates major character alignment drift.

PERFORMANCE TRADE-OFFS

# Latency & Throughput Trade-offs

**Tesseract + PyMuPDF4LLM**
Fastest Processing
**0.3731s**

**PyMuPDF4LLM forced OCR**
Sub-second Execution
**0.4737s**

**MinerU standard**
Heavy Model Execution
**5.7057s**

## 12–15× Latency Overhead

MinerU achieves drastic quality gains but requires significant compute time.

**Decision Rule:** Use lightweight OCR pipelines strictly when real-time throughput dominates. Use MinerU when structural fidelity is critical.

SUMMARY FINDINGS

# Key Benchmark Conclusions

**Best Accuracy: MinerU Standard**
Dominates both structural integrity (0.7389 TEDS-Struct) and text accuracy (0.1644 CER).

**Fastest Throughput: Tesseract + PyMuPDF4LLM**
Achieves 0.3731s per sample but severely compromises table structural quality.

**Best Lightweight Option: Direct Forced OCR**
Direct PyMuPDF4LLM forced OCR outperforms the searchable OCR PDF pipeline across all quality metrics.

**Pipeline Warning: Intermediate OCR PDFs**
Generating searchable OCR PDFs before parsing offers zero benefit and degrades structural fidelity.

ROADMAP

# Limitations & Next Steps

## ⚠ Benchmark Limitations

**Sample Size:** 50 samples; requires scaling for dataset generalization.

**Hardware Variance:** Overhead influenced by environment & hosting setup.

**Conversion Sensitivity:** TEDS penalties for unformed HTML tables.

## Actionable Next Steps

**Scale Evaluation:** Expand benchmark sample size to 500+ tables.

**Failure Audits:** Manually inspect low-scoring table conversions.

**Optimize Latency:** Investigate batching to lower MinerU processing time.